

# Use Case Specification: Pay Order

## 1. Use Case Code

UC002

## 2. Brief Description

This use case describes the interactions between the Customer, AIMS software, and VietQR when the customer desires to pay for a placed order using a QR code. AIMS software generates a VietQR QR code for the customer to scan and complete the payment via their banking application. VietQR then sends a payment callback to AIMS software to confirm the transaction result. This use case is included by the use case “**Place Order**”. If the customer does not wish to pay by QR code, they may switch to credit card payment, which triggers the use case “**Pay Order via PayPal**”.

## 3. Actors

### 3.1. Customer

The customer is the end-user who scans the QR code using their banking application to complete the payment, or who may choose to switch to credit card payment via PayPal.

### 3.2. VietQR

VietQR is an external payment service provider. AIMS software calls VietQR’s API to generate the QR code. VietQR then sends a payment callback to AIMS software with the transaction result.

## 4. Preconditions

- The use case “Place Order” has been initiated and the customer has confirmed the invoice.
- The invoice information — including the list of products in the cart, quantities, product prices, total product price excluding VAT, total product price including VAT, delivery fee (not subject to VAT), and total amount to be paid — has been calculated and temporarily saved by AIMS software.
- AIMS software has valid VietQR API credentials configured.

## 5. Basic Flow of Events

**Step 1.** AIMS software calls the VietQR API to generate a QR code, providing the store’s bank account information, the total amount to be paid (from the temporarily saved invoice), and the transfer content corresponding to the order. (See Table B – Input data for QR code generation.)

**Step 2.** VietQR returns the QR code to AIMS software.

**Step 3.** AIMS software displays the payment screen to the customer, showing: the QR code; bank name and account number of the receiving account; transfer content; and the full invoice information including product list (title, quantity, unit price, amount per line), total product price excluding VAT, total product price including VAT, delivery fee (not subject to VAT), and total amount to be paid. (See Table A – Output data of payment screen.) The customer may also choose to switch to credit card payment via PayPal at this step.

**Step 4.** Customer scans the QR code using their banking application and completes the transfer. (Note: In the test/sandbox environment, the customer triggers VietQR's test callback API to simulate a successful payment transaction.)

**Step 5.** VietQR sends a payment callback to AIMS software with the transaction result, including: transaction ID, transaction content, transaction datetime, confirmed amount, and status. (See Table C – Input data from VietQR payment callback.)

**Step 6.** AIMS software verifies that the callback status is “success” and that the confirmed amount matches the total amount on the invoice.

## 6. Alternative Flows

**Table 1** – Alternative flows of events for UC Pay Order

| No | Location    | Condition                                                                           | Action                                                                                                                                                                                                                                                                                     | Resume Location                     |
|----|-------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 1  | At Step 1   | If the customer chooses to pay via PayPal (before the system calls VietQR)          | <ul style="list-style-type: none"> <li>AIMS software inserts the use case “Pay Order via PayPal”.</li> </ul>                                                                                                                                                                               | Use case ends                       |
| 2  | At Step 1   | If VietQR fails to generate the QR code (e.g., network error, service unavailable)  | <ul style="list-style-type: none"> <li>AIMS software notifies the customer that the payment service is temporarily unavailable.</li> <li>The customer may retry or exit the payment process.</li> </ul>                                                                                    | Use case ends                       |
| 3  | At Step 3   | If the customer wishes to adjust the items in the order or the delivery information | <ul style="list-style-type: none"> <li>AIMS software returns the customer to the delivery information / cart screen.</li> <li>The customer updates the order or delivery information.</li> <li>AIMS software recalculates the delivery fee and updates the invoice accordingly.</li> </ul> | Resumes at Step 8 of UC Place Order |
| 4  | At Step 4   | If the payment transaction is reported as failed by VietQR callback                 | <ul style="list-style-type: none"> <li>AIMS software notifies the customer that the payment was unsuccessful.</li> <li>The customer may retry the payment.</li> </ul>                                                                                                                      | At Step 3                           |
| 5  | At any step | If the customer chooses to go back or exit during the payment process               | <ul style="list-style-type: none"> <li>AIMS software returns the customer to the invoice screen.</li> <li>No payment is recorded.</li> </ul>                                                                                                                                               | Resumes at Step 8 of UC Place Order |

## 7. Input Data

### 7.1. Input data for QR code generation request (AIMS to VietQR)

**Table B** – Input data for VietQR QR code generation

| N<br>o | Data Field   | Description                                              | Mandat<br>ory | Valid<br>Condition                  | Example                           |
|--------|--------------|----------------------------------------------------------|---------------|-------------------------------------|-----------------------------------|
| 1      | bank_id      | Bank identifier of the receiving account                 | Yes           | Valid VietQR bank code              | 970422                            |
| 2      | account_no   | Receiving bank account number                            | Yes           | Non-empty numeric string            | 0123456789                        |
| 3      | amount       | Total amount to be paid (VND), equal to invoice total    | Yes           | Positive integer                    | 123456                            |
| 4      | add_info     | Transfer description / transaction content for the order | Yes           | Non-empty string, max 25 characters | AIMS Order #00123                 |
| 5      | callback_url | URL where VietQR will send the payment result callback   | Yes           | Valid HTTPS URL                     | https://aims.example.com/callback |

## 7.2. Input data from VietQR payment callback (VietQR to AIMS)

**Table C** – Input data from VietQR payment callback

| N<br>o | Data Field           | Description                                      | Mandat<br>ory | Valid Condition                            | Example             |
|--------|----------------------|--------------------------------------------------|---------------|--------------------------------------------|---------------------|
| 1      | transaction_id       | Unique transaction identifier assigned by VietQR | Yes           | Non-empty string                           | TXN20231005001      |
| 2      | transaction_content  | Transfer description confirmed by VietQR         | Yes           | Non-empty string                           | AIMS Order #00123   |
| 3      | transaction_datetime | Date and time the transaction was completed      | Yes           | Valid datetime, format dd/mm/yyyy hh:mm:ss | 05/10/2023 14:30:00 |
| 4      | amount               | Amount received and confirmed by VietQR          | Yes           | Positive integer, must match invoice total | 123456              |
| 5      | status               | Result status of the payment transaction         | Yes           | "success" or "failed"                      | success             |

## 8. Output Data

### 8.1. Output data of payment screen (AIMS to Customer)

**Table A** – Output data of payment screen

| N<br>o | Data Field    | Description                                                | Display Format           | Example    |
|--------|---------------|------------------------------------------------------------|--------------------------|------------|
| 1      | QR Code Image | QR code image generated by VietQR for the customer to scan | Image rendered on screen | [QR image] |
| 2      | Bank name     | Name of the receiving bank                                 | String                   | MB Bank    |

| N<br>o | Data Field                      | Description                                                             | Display Format                                             | Example           |
|--------|---------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------|-------------------|
| 3      | Account number                  | Receiving bank account number                                           | Numeric string                                             | 0123456789        |
| 4      | Transfer content                | Transfer description corresponding to the order                         | String                                                     | AIMS Order #00123 |
| 5      | Product title                   | Title of each media product in the order                                | String                                                     | DVD Mat Biec      |
| 6      | Quantity                        | Quantity of each product in the order                                   | Positive integer, right-aligned                            | 2                 |
| 7      | Unit price                      | Price per unit of each product (excluding VAT)                          | Comma thousands separator, positive integer, right-aligned | 1,234,500         |
| 8      | Amount                          | Total per product line (unit price × quantity, excl. VAT)               | Comma thousands separator, positive integer, right-aligned | 1,234,500         |
| 9      | Total product price (excl. VAT) | Sum of all product amounts before VAT                                   | Comma thousands separator, right-aligned                   | 1,234,500         |
| 10     | Total product price (incl. VAT) | Total product price with 10% VAT applied                                | Comma thousands separator, right-aligned                   | 1,234,500         |
| 11     | Delivery fee                    | Shipping fee based on weight and delivery location. Not subject to VAT. | Comma thousands separator, right-aligned                   | 36,360            |
| 12     | Total amount to be paid         | Total = total product price (incl. VAT) + delivery fee                  | Comma thousands separator, right-aligned                   | 1,234,500         |

## 9. Postconditions

### 9.1. Success

- The payment transaction is confirmed as successful.
- AIMS software records the payment transaction information (transaction ID, transaction content, transaction datetime) and the successfully paid order.
- The order is placed in the “pending processing” state, awaiting review by the product manager.
- An email containing the invoice and payment transaction information is sent to the customer.

### 9.2. Failure / Cancellation

- If payment is unsuccessful or the customer exits during the payment process, no payment record is created and no order is saved.
- Control returns to the use case “Place Order” at the invoice screen, allowing the customer to retry or go back.
- Note: VietQR does not support an automated refund API. Therefore, if an order that was successfully paid via VietQR is subsequently cancelled (by the customer or

rejected by the product manager), no automated refund will be triggered. In such cases, the product manager is responsible for issuing the refund manually (e.g., by contacting the customer and transferring funds outside the system). See the use case **“Cancel Order”** for the full cancellation flow.